

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12403650
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Ostroverkhov; Victor Petrovich et al.

Additive manufacturing methods and systems

Abstract

Methods of additively manufacturing a three-dimensional object include irradiating a first build plane region using a first energy beam defining a beam diameter, the first energy beam travelling along a first oscillating path in a first direction to consolidate a first wall defining a thickness perpendicular to the first direction, wherein a build material adjacent a first side of the first wall and the build material adjacent a second side of the first wall, opposite the first side of the first wall, remains unconsolidated; and wherein the thickness of the first wall is greater than the beam diameter.

Inventors: Ostroverkhov; Victor Petrovich (Ballston Lake, NY), Immer; Christopher Darby (Niskayuna, NY), Adcock; Thomas Charles (Glenville, NY), Gambone; Justin John (Schenectady, NY), Erno; Daniel Jason (Clifton Park, NY), McCarthy; Brian Scott (Clifton Park, NY), Madelone, Jr.; John Joseph (South Glen Falls, NY)

Applicant: General Electric Company (Schenectady, NY)

Family ID: 87245429

Assignee: General Electric Company (Evendale, OH)

Appl. No.: 17/865489

Filed: July 15, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20240017481 A1	Jan. 18, 2024

Publication Classification

Int. Cl.: B29C64/153 (20170101); **B22F10/28** (20210101); **B22F12/45** (20210101); **B28B1/00** (20060101); **B29C64/268** (20170101); **B29C64/282** (20170101); **B33Y10/00** (20150101); **B33Y30/00** (20150101)

U.S. Cl.:

CPC B29C64/153 (20170801); **B22F10/28** (20210101); **B22F12/45** (20210101); **B28B1/001** (20130101); **B29C64/268** (20170801); **B29C64/282** (20170801); **B33Y10/00** (20141201); **B33Y30/00** (20141201);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4917484	12/1989	Heinz	N/A	N/A
5550669	12/1995	Patel	N/A	N/A
6756561	12/2003	McGregor et al.	N/A	N/A
6807892	12/2003	Biegelsen et al.	N/A	N/A
7188831	12/2006	Coppoolse	N/A	N/A
7899569	12/2010	Holzwarth	N/A	N/A
8034279	12/2010	Dimter et al.	N/A	N/A
8752969	12/2013	Kane et al.	N/A	N/A
9527246	12/2015	Wiesner et al.	N/A	N/A
9878497	12/2017	Schwarze et al.	N/A	N/A
10035188	12/2017	Weilhammer et al.	N/A	N/A
10337335	12/2018	Pavlov et al.	N/A	N/A
10372110	12/2018	Jagdale et al.	N/A	N/A
10399183	12/2018	Dallarosa et al.	N/A	N/A
10406633	12/2018	Crear et al.	N/A	N/A
10471665	12/2018	Marshall, Jr. et al.	N/A	N/A
10661514	12/2019	Talgorn et al.	N/A	N/A
10751835	12/2019	Grapov et al.	N/A	N/A
10786865	12/2019	Hellestam	N/A	N/A
10814429	12/2019	Roychowdhury et al.	N/A	N/A
10821551	12/2019	Roychowdhury et al.	N/A	N/A
10912211	12/2020	Range et al.	N/A	N/A
11072113	12/2020	Zeulner	N/A	N/A
11135680	12/2020	Wuest et al.	N/A	N/A
11161201	12/2020	Ostroverkhov et al.	N/A	N/A
11226003	12/2021	Koczar et al.	N/A	N/A
11344978	12/2021	DeMuth et al.	N/A	N/A
2011/0311389	12/2010	Ryan et al.	N/A	N/A
2012/0133080	12/2011	Moussa et al.	N/A	N/A

2012/0267347	12/2011	Arjakine et al.	N/A	N/A
2013/0216836	12/2012	Grebe et al.	N/A	N/A
2013/0263977	12/2012	Rickenbacher et al.	N/A	N/A
2014/0140882	12/2013	Syassen	N/A	N/A
2014/0348691	12/2013	Ljungblad et al.	N/A	N/A
2015/0198052	12/2014	Pavlov	416/241B	F01D 5/284
2015/0210013	12/2014	Teulet	N/A	N/A
2017/0129185	12/2016	Buller	N/A	B23K 26/702
2017/0173883	12/2016	Gray et al.	N/A	N/A
2017/0197278	12/2016	Garry et al.	N/A	N/A
2017/0304894	12/2016	Buller	N/A	N/A
2018/0099331	12/2017	Hoebel et al.	N/A	N/A
2018/0178287	12/2017	Mamrak	N/A	B23K 26/342
2018/0207722	12/2017	Feldmann et al.	N/A	N/A
2018/0281284	12/2017	Elgar	N/A	B29C 64/371
2018/0304406	12/2017	Roerig et al.	N/A	N/A
2018/0326536	12/2017	Mentzel et al.	N/A	N/A
2018/0345405	12/2017	Ostroverkhov	N/A	B33Y 50/02
2018/0345413	12/2017	Wuest	N/A	B33Y 30/00
2019/0017757	12/2018	Seeholzer et al.	N/A	N/A
2019/0091935	12/2018	Weichsellbaumer et al.	N/A	N/A
2019/0217542	12/2018	Pontiller-Schymura et al.	N/A	N/A
2019/0232427	12/2018	Roychowdhury et al.	N/A	N/A
2019/0232428	12/2018	Roychowdhury et al.	N/A	N/A
2019/0291184	12/2018	Buller	N/A	B29C 64/20
2020/0223139	12/2019	Hunze et al.	N/A	N/A
2020/0276667	12/2019	Filkins et al.	N/A	N/A
2020/0353678	12/2019	Yeoh	N/A	N/A
2021/0187830	12/2020	McCarthy et al.	N/A	N/A
2021/0316409	12/2020	DeMuth et al.	N/A	N/A
2022/0168813	12/2021	Dardis et al.	N/A	N/A
2022/0176452	12/2021	De Chirico	N/A	B29C 64/268

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
112496339	12/2020	CN	N/A
102017200793	12/2017	DE	N/A

WO-2016196382	12/2015	WO	B22F 3/1055
WO-2021229171	12/2020	WO	B22F 10/28
WO2021229172	12/2020	WO	N/A

OTHER PUBLICATIONS

Blanchet E. “English Machine Translation of WO 2021229171 A1: Method for the Additive Manufacture of an Object From a Powder Layer”. Espacenet.EPO. 2023 (Year: 2023). cited by examiner

Shi et al., Microstructural Control in Metal Laser Powder Bed Fusion Additive Manufacturing Using Laser Beam Shaping Strategy, Acta Materialia, vol. 184, Feb. 1, 2020, pp. 284-305.

<https://doi.org/10.1016/j.actamat.2019.11.053>. cited by applicant

SLM Solutions, NXG XII 600, 10 Pages Retrieved Jan. 5, 2022 from <https://www.slm-pushing-the-limits.com/specs#optical-system>. cited by applicant

Tenbrock et al., Influence of Keyhole and Conduction Mode Melting for Top-Hat Shaped Beam Profiles in Laser Powder Bed Fusion, Journal of Materials Processing Technology, vol. 278, Apr. 2020, 10 Pages. <https://doi.org/10.1016/j.imatprotec.2019.116514>. cited by applicant

Tumkur et al., Nondiffractive Beam Shaping for Enhanced Optothermal Control in Metal Additive Manufacturing, Science Advances, vol. 17, Issue 38, Sep. 15, 2021, 12 Pages.

<https://www.science.org/doi/pdf/10.1126/sciadv.abg9358>. cited by applicant

Cen et al., “Oscillating Laser-Arc Hybrid Additive Manufacturing of AZ31 Magnesium Alloy”, Proceedings of the SPIE, vol. 12312, Dec. 19, 2022, 15 pages. cited by applicant

Gao et al., “Defect elimination and microstructure improvement of laser powder bed fusion β -solidifying γ -TiAl alloys via circular beam oscillation technology”, Materials Science, Elsevier, Amsterdam, vol. 873, Apr. 1, 2023 (Abstract Only). cited by applicant

Primary Examiner: Leong; Susan D

Assistant Examiner: Woo; Jonathan B

Attorney, Agent or Firm: Dinsmore & Shohl LLP

Background/Summary

FIELD

(1) The present disclosure relates to additive manufacturing methods and systems, such as additive manufacturing methods and systems to build thin walls.

BACKGROUND

(2) Three-dimensional objects may be additively manufactured using a variety of methods and systems. For example, additive manufacturing may involve a powder bed fusion process in which one or more energy beams are directed onto a powder bed to melt, fuse, or sinter sequential layers of build material such as powder material. The properties of the three-dimensional object formed by consolidating the powder material depend, at least in part, on one or more parameters of the energy beam. Additionally, one or more parameters of an energy beam impact operating parameters such as processing speed of the additive manufacturing process.

(3) In some additive manufacturing systems, one or more walls may be built using one or more energy beams. For example, a wall is built by consolidating outer edges of the wall with one or more passes by one or more energy beams. The internal portion of the wall can further be consolidated by using additional passes parallel to the edges or by using a distinct hatching pattern

to otherwise fill the interior of the wall. The one or more additively manufactured walls can include structures that define, for example, an outer surface or an internal feature of the three-dimensional object, or even provide a lattice support structure to the interior of the three-dimensional object.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) A full and enabling disclosure of the present disclosure, including the best mode thereof, directed to one of ordinary skill in the art, is set forth in the specification, which makes reference to the appended figures, in which:
- (2) FIG. 1 schematically depicts an exemplary additive manufacturing system in accordance with one or more exemplary aspects of the present disclosure.
- (3) FIG. 2 schematically depicts a build plane in accordance with one or more exemplary aspects of the present disclosure.
- (4) FIG. 3 schematically depicts an oscillation of an energy beam in accordance with one or more exemplary aspects of the present disclosure.
- (5) FIG. 4 schematically depicts an energy beam having a first oscillating path forming a first wall in accordance with one or more exemplary aspects of the present disclosure.
- (6) FIG. 5 schematically depicts an energy beam having a first oscillating path with variations forming another first wall in accordance with one or more exemplary aspects of the present disclosure.
- (7) FIG. 6 schematically depicts an energy beam having multiple oscillating paths forming yet another first wall in accordance with one or more exemplary aspects of the present disclosure.
- (8) FIG. 7 schematically depicts multiple energy beams having multiple oscillating paths to form yet another first wall and an extension in accordance with one or more exemplary aspects of the present disclosure.
- (9) FIG. 8 schematically depicts multiple energy beams having multiple oscillating paths with variations to form yet another first wall and an extension in accordance with one or more exemplary aspects of the present disclosure.
- (10) FIG. 9 schematically depicts multiple energy beams having multiple tapered oscillating paths to form yet another first wall and an extension in accordance with one or more exemplary aspects of the present disclosure.
- (11) FIG. 10 schematically depicts multiple energy beams having multiple tapered oscillating paths to form yet another first wall and an extension in accordance with one or more exemplary aspects of the present disclosure.
- (12) FIG. 11 schematically depicts a three-dimensional object having multiple intersecting walls in accordance with one or more exemplary aspects of the present disclosure.
- (13) FIG. 12 schematically depicts an exemplary method of additively manufacturing a three-dimensional object in accordance with one or more exemplary aspects of the present disclosure.
- (14) FIG. 13 schematically depicts an exemplary controls system that may be configured to control an additive manufacturing system or machine in accordance with one or more exemplary aspects of the present disclosure.

DETAILED DESCRIPTION

- (15) Reference will now be made in detail to present embodiments of the disclosure, one or more examples of which are illustrated in the accompanying drawings. The detailed description uses numerical and letter designations to refer to features in the drawings. Like or similar designations in the drawings and description have been used to refer to like or similar parts of the disclosure.
- (16) As described herein, the presently disclosed subject matter involves the use of additive manufacturing machines or systems. As used herein, the term “additive manufacturing” refers

generally to manufacturing technology in which components are manufactured in a layer-by-layer manner. An exemplary additive manufacturing machine may be configured to utilize any suitable additive manufacturing technology. The additive manufacturing machine may utilize an additive manufacturing technology that includes a powder bed fusion (PBF) technology, such as a direct metal laser melting (DMLM) technology, an electron beam melting (EBM) technology, a selective laser melting (SLM) technology, a directed metal laser sintering (DMLS) technology, or a selective laser sintering (SLS) technology. In an exemplary PBF technology, thin layers of powder material are sequentially applied to a build plane and then selectively melted or fused to one another in a layer-by-layer manner to form one or more three-dimensional objects. Additively manufactured objects are generally monolithic in nature and may have a variety of integral sub-components.

(17) Additionally or alternatively suitable additive manufacturing technologies may include, for example, Fused Deposition Modeling (FDM) technology, Direct Energy Deposition (DED) technology, Laser Engineered Net Shaping (LENS) technology, Laser Net Shape Manufacturing (LNSM) technology, Direct Metal Deposition (DMD) technology, Digital Light Processing (DLP) technology, and other additive manufacturing technologies that utilize an energy beam or other energy source to solidify an additive manufacturing material such as a powder material. In fact, any suitable additive manufacturing modality may be utilized with the presently disclosed the subject matter.

(18) Additive manufacturing technology may generally be described as fabrication of objects by building objects point-by-point, line-by-line, layer-by-layer, typically in a vertical direction. Other methods of fabrication are contemplated and within the scope of the present disclosure. For example, although the discussion herein refers to the addition of material to form successive layers, the presently disclosed subject matter may be practiced with any additive manufacturing technology or other manufacturing technology, including layer-additive processes, layer-subtractive processes, or hybrid processes.

(19) The additive manufacturing processes described herein may be used for forming components using any suitable material. For example, the material may be metal, ceramic, polymer, epoxy, photopolymer resin, plastic, or any other suitable material that may be in solid, powder, sheet material, wire, or any other suitable form, or combinations thereof. Additionally, or in the alternative, exemplary materials may include metals, ceramics, or binders, as well as combinations thereof. Exemplary ceramics may include ultra-high-temperature ceramics, or precursors for ultra-high-temperature ceramics, such as polymeric precursors. Each successive layer may be, for example, between about 10 μm and 200 μm , although the thickness may be determined based on any number of parameters and may be any suitable size. Moreover, the additive manufacturing process described herein may be used for forming any type of suitable component. For example, the component formed using the additive manufacturing process described herein may comprise one or more turbine components such as turbine blades, shrouds, nozzles, heat shields, or vanes.

(20) As used herein, the term “build plane” refers to a plane defined by a surface upon which an energy beam impinges to selectively irradiate and thereby consolidate powder material during an additive manufacturing process. Generally, the surface of a powder bed defines the build plane. During irradiation of a respective layer of the powder bed, a previously irradiated portion of the respective layer may define a portion of the build plane. Prior to distributing powder material across a build module, a build plate that supports the powder bed generally defines the build plane.

(21) As used herein, the term “consolidate” or “consolidating” refers to solidification of build material (e.g., powder material) as a result of irradiating the build material, including by way of melting, fusing, sintering, or the like such that multiple separate pieces of build material (e.g., multiple individual pieces of powder material) join together into a single structure.

(22) As used herein, the term “unconsolidated” refers to separate pieces of material than are not bonded or otherwise joined to one another, such as separate pieces of loose powder.

(23) It is also understood that terms such as “top”, “bottom”, “outward”, “inward”, and the like are

words of convenience and are not to be construed as limiting terms.

(24) As used herein, the terms “first”, “second”, and “third” may be used interchangeably to distinguish one component from another and are not intended to signify location or importance of the individual components.

(25) The terms “a” and “an” do not denote a limitation of quantity, but rather denote the presence of at least one of the referenced item.

(26) Approximating language, as used herein throughout the specification and claims, may be applied to modify any quantitative representation that could permissibly vary without resulting in a change in the basic function to which it is related. Accordingly, a value modified by a term or terms, such as “about,” “substantially,” and “approximately,” are not to be limited to the precise value specified. In at least some instances, the approximating language may correspond to the precision of an instrument for measuring the value, or the precision of the methods or machines for constructing or manufacturing the components or systems. For example, the approximating language may refer to being within a 10 percent margin.

(27) Here and throughout the specification and claims, range limitations are combined and interchanged, such ranges are identified and include all the sub-ranges contained therein unless context or language indicates otherwise. For example, all ranges disclosed herein are inclusive of the endpoints, and the endpoints are independently combinable with each other.

(28) The present disclosure is generally related to additive manufacturing methods and systems to build thin walls for three-dimensional objects. One or more energy beams in such systems can be used to build walls to define, for example, an outer surface, an internal feature, or an internal support lattice of a three-dimensional object. However, the minimal obtainable thickness of the wall may be inhibited by requiring multiple passes of an energy beam. Moreover, the uniformity of the consolidated build material within the wall may be inhibited when utilizing multiple passes or multiple energy beams.

(29) Accordingly, alternative additive manufacturing methods and systems would be welcomed in the art, including additive manufacturing methods and systems that provide thin walls using energy beam oscillations.

(30) The presently disclosed subject matter will now be described in further detail. FIG. 1 schematically depicts an additive manufacturing system **100**. The additive manufacturing system **100** may include one or more additive manufacturing machines **102**. It will be appreciated that the additive manufacturing system **100** and additive manufacturing machine **102** shown in FIG. 1 is provided by way of example and not to be limiting. In fact, the subject matter of the present disclosure may be practiced with any additive manufacturing system **100** and additive manufacturing machine **102** without departing from the scope of the present disclosure. As shown, the one or more additive manufacturing machines **102** may include a control system **104**. The control system **104** may be included as part of the additive manufacturing machine **102** or the control system **104** may be associated with the additive manufacturing machine **102**. The control system **104** may include componentry integrated as part of the additive manufacturing machine **102** or componentry that is provided separately from the additive manufacturing machine **102**. Various componentry of the control system **104** may be communicatively coupled to various componentry of the additive manufacturing machine **102**.

(31) The control system **104** may be communicatively coupled with a management system **106** or a user interface **108**. The management system **106** may be configured to interact with the control system **104** in connection with enterprise-level operations pertaining to the additive manufacturing system **100**. Such enterprise level operations may include transmitting data from the management system **106** to the control system **104** or transmitting data from the control system **104** to the management system **106**. The user interface **108** may include one or more user input/output devices to allow a user to interact with the additive manufacturing system **100**.

(32) As shown, an additive manufacturing machine **102** may include a build module **110** that

includes a build chamber **112** within which an object or three-dimensional objects **114** may be additively manufactured. An additive manufacturing machine **102** may include a powder module **116** that contains a supply of build material **118** (e.g., powder material) housed within a supply chamber **120**. The build module **110** or the powder module **116** may be provided in the form of modular containers configured to be installed into and removed from the additive manufacturing machine **102** such as in an assembly-line process. Additionally, or in the alternative, the build module **110** or the powder module **116** may define a fixed componentry of the additive manufacturing machine **102**.

(33) The powder module **116** contains a supply of build material **118** housed within a supply chamber **120**. The powder module **116** includes a powder piston **122** that elevates a powder floor **124** during operation of the additive manufacturing machine **102**. As the powder floor **124** elevates, a portion of the build material **118** is forced out of the powder module **116**. A recoater **126** such as a blade or roller sequentially distributes thin layers of build material **118** across a build plane **128** above the build module **110**. A build platform **130** supports the sequential layers of build material **118** distributed across the build plane **128**. A build platform **130** may include a build plate (not shown) secured thereto and upon which a three-dimensional object **114** may be additively manufactured.

(34) The additive manufacturing machine **102** includes an energy beam system **132** configured to generate one or more energy beams **134** and to direct the energy beams **134** onto the build plane **128** to selectively solidify respective portions of the powder bed **136** defining the build plane **128**. The energy beams **134** may be laser beams or beams from any other suitable energy source, such as LEDs or other light sources, and so forth. As the energy beams **134** selectively melt or fuse the sequential layers of build material **118** that define the powder bed **136**, the three-dimensional object **114** begins to take shape. The one or more energy beams **134** or laser beams may include electromagnetic radiation having any suitable wavelength or wavelength range, such as a wavelength or wavelength range corresponding to infrared light, visible light, or ultraviolet light, or a combination thereof.

(35) Typically, with a DMLM, EBM, or SLM system, the build material **118** is fully melted, with respective layers being melted or re-melted with respective passes of the energy beams **134**. With DMLS or SLS systems, typically the layers of build material **118** are sintered, fusing particles of build material **118** to one another generally without reaching the melting point of the build material **118**. The energy beam system **132** may include componentry integrated as part of the additive manufacturing machine **102** or componentry that is provided separately from the additive manufacturing machine **102**.

(36) The energy beam system **132** may include one or more irradiation devices **138** configured to generate a plurality of energy beams **134** and to direct the energy beams upon the build plane **128**. An energy beam system **132** may include a plurality of irradiation devices **138**, such as a first irradiation device **138a** and a second irradiation device **138b**. The one or more irradiation devices **138** may respectively include an energy beam source **140** (e.g., first energy beam source **140a** and second energy beam source **140b**), an optical assembly **142** (e.g., first optical assembly **142a** and second optical assembly **142b**), and a scanner **144** (e.g., first scanner **144a** and second scanner **144b**). The optical assembly **142** may include a plurality of optical elements configured to direct the energy beam onto the build plane **128**. The optical assembly **142** may include one or more optical elements, such as lenses through which an energy beam may be transmitted along an optical path from the energy beam source to the build plane. By way of example, an optical assembly **142** may include one more focusing lenses that focus an energy beam **134** on a build plane **128**. A scanner **144** may include a galvo scanner, an electro-optic modulator, an acousto-optic modulator, a piezo-driven mirror, or the like. Additionally, or in the alternative, the energy beam system **132** may include a window **146**, such as a protective glass, that separates one or more components of the energy beam system **132** from the environment of a process chamber **148** within which build

material **118** is irradiated by the one or more energy beams **134** to additively manufacture a three-dimensional object **114**.

(37) The window **146** may prevent contaminants from fumes associated with the additive manufacturing process, such as powder material, dust, soot, residues, vapor, byproducts, and the like, from coming into contact with sensitive components of an energy beam system **132**.

Accumulation of contaminants upon various optical elements of an optical assembly **142** may adversely affect operation of the energy beam system **132** or quality metrics associated with an energy beam system. Additionally, or in the alternative, such contaminants may cause damage to various optical elements of an optical assembly **142**.

(38) As shown in FIG. 1, the energy beam system **132** includes a first irradiation device **138a** and a second irradiation device **138b**. Additionally, or in the alternative, an energy beam system **132** may include any number of additional irradiation devices such as three, four, six, eight, ten, or more irradiation devices, and such irradiation devices may respectively include an optical assembly **142**. The plurality of irradiation devices **138** may be configured to respectively generate one or more energy beams that are respectively scannable within a scan field incident upon at least a portion of the build plane **128** to selectively consolidate the portions of the build material **118** that are to become part of the three-dimensional object **114**.

(39) For example, the first irradiation device **138a** may generate a first energy beam **134a** that is scannable within a first scan field **150a** incident upon at least a first build plane region **152a**. The second irradiation device **138b** may generate a second energy beam **134b** that is scannable within a second scan field **150b** incident upon at least a second build plane region **152b**. The first scan field **150a** and the second scan field **150b** may overlap such that the first build plane region **152a** scannable by the first energy beam **134a** overlaps with the second build plane region **152b** scannable by the second energy beam **134b**. The overlapping portion of the first build plane region **152a** and the second build plane region **152b** may sometimes be referred to as an interlace region **154**. Portions of the powder bed **136** to be irradiated within the interlace region **154** may be irradiated by the first energy beam **134a**, or the second energy beam **134b**, or a combination thereof. While the powder bed **136** to be irradiated is exemplary illustrated as being irradiated by the first energy beam **134a** or the second energy beam **134b**, it is appreciated that any number of energy beams **134** may additionally or alternatively be utilized for irradiating the build plane **128** in accordance with the present disclosure.

(40) To irradiate a layer of the powder bed **136**, the one or more irradiation devices **138** (e.g., the first irradiation device **138a** and the second irradiation device **138b**) respectively direct the plurality of energy beams **134** beams (e.g., the first energy beam **134a** and the second energy beam **134b**) across the respective portions of the build plane **128** (e.g., the first build plane region **152a** and the second build plane region **152b**) to selectively consolidate the portions of the build material **118** that are to become part of the three-dimensional object **114**. The one or more energy beams **134** may become incident upon the build plane **128** defined by the powder bed **136**, for example, after passing through one or more optical elements of the optical assembly **142** or through a window **146** of the energy beam system **132**. As sequential layers of the powder bed **136** are consolidated, a build piston **156** gradually lowers the build platform **130** to make room for sequential layers of build material **118**. As sequential layers of build material **118** are applied across the build plane **128**, the next sequential layer of build material **118** defines the surface of the powder bed **136** coinciding with the build plane **128**. Sequential layers of the powder bed **136** may be selectively consolidated until a completed object **114** has been additively manufactured. In some aspects of the disclosure, an additive manufacturing machine may utilize an overflow module (not shown) to capture excess build material **118**. Additionally, or in the alternative, excess build material **118** may be redistributed across the build plane **128** when applying a next sequential layer of build material **118**. It will be appreciated that other systems may be provided for handling the build material **118**, including different powder supply systems or excess powder recapture systems. The subject matter

of the present disclosure may be practiced with any suitable additive manufacturing machine without departing from the scope hereof.

(41) Still referring to FIG. 1, an additive manufacturing machine **102** may include an imaging system **158** (e.g., first imaging system **158a** and second imaging system **158b**) configured to monitor one or more operating parameters of an additive manufacturing machine **102**, one or more parameters of an energy beam system **132**, or one or more operating parameters of an additive manufacturing process. The imaging system may have a calibration system configured to calibrate one or more operating parameters of an additive manufacturing machine **102** or of an additive manufacturing process. The imaging system **158** may be a melt pool monitoring system. The one or more operating parameters of the additive manufacturing process may include operating parameters associated with additively manufacturing a three-dimensional object **114**. The imaging system **158** may be configured to detect an imaging beam such as an infrared beam from a laser diode or a reflected portion of an energy beam (e.g., a first energy beam **134a** or a second energy beam **134b**).

(42) An energy beam system **132** or an imaging system **158** may include one or more detection devices. The one or more detection devices may be configured to determine one or more parameters of an energy beam system **132**, such as one or more parameters associated with irradiating the sequential layers of the powder bed **136** based at least in part on an assessment beam detected by the imaging system **158**. One or more parameters associated with consolidating the sequential layers of the powder bed **136** may include irradiation parameters or object parameters, such as melt pool monitoring parameters. The one or more parameters determined by the imaging system **158** may be utilized, for example, by the control system **104**, to control one or more operations of the additive manufacturing machine **102** or of the additive manufacturing system **100**. The one or more detection devices may be configured to obtain assessment data of the build plane **128** from a respective assessment beam. An exemplary detection device may include a camera, an image sensor, a photo diode assembly, or the like. For example, a detection device may include charge-coupled device (e.g., a CCD sensor), an active-pixel sensor (e.g., a complementary metal-oxide semiconductor (CMOS) sensor), a quanta image device (e.g., a QIS sensor), or the like. A detection device may additionally include a lens assembly configured to focus an assessment beam along a beam path to the detection device. An imaging system **158** may include one or more imaging optical elements (not shown), such as mirrors, beam splitters, lenses, and the like, configured to direct an assessment beam to a corresponding detection device.

(43) In addition, or in the alternative, to determine parameters associated with irradiation the sequential layers of the powder bed **136**, the imaging system **158** may be configured to perform one or more calibration operations associated with an additive manufacturing machine **102**, such as a calibration operation associated with the energy beam system **132**, one or more irradiation devices **138** or components thereof, or the imaging system **158** or components thereof. The imaging system **158** may be configured to project an assessment beam and to detect a portion of the assessment beam reflected from the build plane **128**. The assessment beam may be projected by an irradiation device **138** or a separate beam source associated with the imaging system **158**. Additionally, or in the alternative, the imaging system **158** may be configured to detect an assessment beam that includes radiation emitted from the build plane **128**, such as radiation from an energy beam **134** reflected from the powder bed **136** or radiation emitted from a melt pool in the powder bed **136** generated by an energy beam **134** or radiation emitted from a portion of the powder bed **136** adjacent to the melt pool. The imaging system **158** may include componentry integrated as part of the additive manufacturing machine **102** or componentry that is provided separately from the additive manufacturing machine **102**. For example, the imaging system **158** may include componentry integrated as part of the energy beam system **132**. Additionally, or in the alternative, the imaging system **158** may include separate componentry, such as in the form of an assembly, that can be installed as part of the energy beam system **132** or as part of the additive manufacturing machine **102**.

(44) Still referring to FIG. 1, in some aspects of the disclosure, an inertization system **160** may supply a flow of inert process gas **162** to one or more regions the process chamber **148**, such as a region between the energy beam system **132** and the powder bed **136**. The flow of inert process gas **162** may remove fumes from the process chamber **148** or to reduce the tendency of fumes to interfere with the energy beams **134** used to irradiate the build material **118**. Such fumes may present in the form of a plume emanating from a consolidation zone where an energy beam **134** becomes incident upon the powder bed **136** and may sometimes be referred to as a fume plume. A fume plume may include build material, dust, soot, residues, vapors, byproducts, or the like. The flow if inert process gas **162** may also reduce the tendency of contaminants from fumes to deposit on the window **146**, optical elements of the optical assembly **142**, or other components of the energy beam system **132**. The inertization system **160** may provide a directional flow of inert process gas **162** that flows across the build plane **128**. For example, as shown, the inert process gas **162** flows from left to right. The inertization system **160** may include a supply manifold **164** and a return manifold **166**. The inert process gas **162** may flow from the supply manifold **164** to the return manifold **166**. Fumes in the process chamber **148** may be drawn into the return manifold **166**. In some aspects of the disclosure, the supply manifold **164** or the return manifold **166** may be coupled to, or define a portion of, a perimeter wall of the process chamber **148**. Additionally, or in the alternative, the supply manifold **164** or the return manifold **166** may be coupled to a housing assembly **168** that contains one or more components of the energy beam system **132**, such as one or more irradiation devices **138** and or one or more imaging systems **158**. With the supply manifold **164** or the return manifold **166** coupled to the housing assembly **168**, a relatively small volume of space between the energy beam system **132** and the powder bed **136** may be inertized, as opposed to inertizing an entire interior of the process chamber **148**. Additionally, or in the alternative, a fume plume may have a shorter path to travel before being drawn into the return manifold **166** by the flow of inert process gas **162**.

(45) The energy beam system **132** may be positioned at any suitable location within the process chamber **148**. Additionally, or in the alternative, the energy beam system **132** may be coupled to a perimeter wall of the process chamber **148**. In some aspects of the disclosure, an additive manufacturing machine may include a positioning system **170** configured to move an energy beam system **132** or one or more components thereof relative to the build plane **128**. The positioning system **170** may be configured to move the energy beam system **132** or one or more components thereof to specified build coordinates or along specified build vectors corresponding to a cartesian coordinate system in accordance with control commands provided, for example, by the control system **104**. The control commands may be provided, for example, to carry out operations of the one or more energy beam system **132** or of the additive manufacturing machine **102** in accordance with the present disclosure. The positioning system **170** may include one or more gantry elements **172** configured to move the energy beam system **132** or one or more components thereof across the powder bed. The gantry elements **172** may respectively be configured to move the energy beam system **132** or one or more components thereof in one or more directions, such as an X-direction, a Y-direction, or a Z-direction. In some aspects of the disclosure, the positioning system **170** may be coupled to the housing assembly **168** that contains one or more components of the energy beam system **132**. The housing assembly **168** may be coupled to one or more gantry elements **172** by one or more gantry mounts **174**. The positioning system **170** may include a drive motor **176** configured to move the housing assembly **168** or the one or more components the energy beam system **132** according to instructions for the control system **104**. The positioning system **170** may include componentry typically associated with a gantry system, such as stepper motors, drive elements, carriages, and so forth.

(46) Referring now additionally to FIG. 2, a top down view of a build plane **128** is illustrated for forming a first wall **153a** of the three-dimensional object **114**. The first wall **153a** can generally refer to any thin or narrow structure consolidated by one or more energy beams **134** (not shown in

FIG. 2). The first wall **153a** extends in a first direction **D1** and can be defined by a first side **157a** of the first wall **153a** and a second side **159a** of the first wall **153a**, wherein the second side **159a** is opposite the first side **157a**. Build material **118** adjacent the first side **157a** of the first wall **153a** and adjacent the second side **159a** of the first wall **153a** remains unconsolidated. That is, the build material **118** adjacent the first wall **153a** remains in its form (e.g., loose powder material) when applying the build material **118** to the build plane **128**.

(47) The first wall **153a** may comprise a variety of configurations and orientations. For example, the first wall **153a** may comprise a generally linear orientation such as illustrated in FIG. 2. In some aspects of the disclosure, the first wall **153a** may comprise a non-linear orientation such that it includes one or more bends, turns, curves or the like. Further, the first wall **153a** comprises a thickness **T** in the thickness direction **DT** which is perpendicular to the first direction **D1** (that is, the distance between the first side **157a** of the first wall **153a** and the second side **159a** of the first wall **153a**). In some aspects of the disclosure, the first wall **153a** may comprise a substantially uniform thickness **T** as it extends in the first direction **D1**. However, in some aspects of the disclosure, the thickness **T** of the first wall **153a** may increase or decrease at one or more locations along the first direction **D1**.

(48) Further, while FIG. 2 illustrates a first wall **153a** in isolation within the build plane **128**, it is appreciated that other structures may also be consolidated in the build plane **128** at one or more locations. For example, the first wall **153a** may extend between and connect to one or more larger structures separated by a defined distance. Alternatively, or additionally, the first wall **153a** may intersect with additional structures in the build plane **128**.

(49) Referring now additionally to FIG. 3, a top down view of a first oscillating path **190** is illustrated for consolidating the first wall **153a** illustrated in FIG. 2. The first oscillating path **190** generally comprises a plurality of oscillations **192** that repeat along a first direction **D1**. That is, the first spot **135a** of the first energy beam **134a** can wobble as it travels across the build plane **128** in the first direction **D1** to produce the plurality of oscillations **192**. The wobbling of the first energy beam **134a** can allow for a larger melt pool in the build plane **128** than if the first spot **135a** were to merely travel linearly in parallel with the first direction **D1**. For instance, the first spot **135a** comprises a beam diameter **DB** defining the diameter of the first spot **135a** on the build plane **128**. The melt pool **MP** formed in the build plane **128** by the plurality of oscillations **192** can exceed the size of the beam diameter **DB** to consolidate a larger area via a single pass or fewer passes. For example, due in part to the plurality of oscillations **192**, the thickness **T** of the first wall **153a** can be greater than the beam diameter **DB**. In some aspects of the disclosure, the thickness **T** of the first wall **153a** may be between 1.5 times and seven times the beam diameter **DB** of the first energy beam **134a**. In some aspects of the disclosure, the thickness **T** of the first wall **153a** may be between two times and five times the beam diameter **DB** of the first energy beam **134a**. In some aspects of the disclosure, the thickness **T** of the first wall **153a** may be between two times and three times the beam diameter **DB** of the first energy beam **134a**.

(50) The first oscillating path **190** can comprise a variety of parameters and configurations including with respect to the plurality of oscillations **192**. For instance, each oscillation **192** may generally comprise an amplitude **A** and a length **L**. The amplitude **A** refers to the maximum distance reached by the first spot **135a** extending away from a midline **M** which bisects the first oscillating path **190** in the first direction **D1**. The length **L** refers to the distance extending in the first direction **D1** of each oscillation **192**. Moreover, the first oscillating path **190** can comprise a distance **D** between oscillations **192**.

(51) The amplitude **A** of each oscillation **192** may impact the size of the overall melt pool, which can have a diameter the same as or similar to the overall oscillation height **OH**. For instance, a greater amplitude can create a greater melt pool size which will, in turn, lead to the consolidation of larger amount of build material **118** (FIGS. 1 and 2) in the build plane **128** from a single pass of the first energy beam **134a**. The length **L** of each oscillation **192** may impact the amount of energy

imparted on the build material **118** (FIGS. **1** and **2**) in the build plane **128** from the first energy beam **134a** by adjusting the amount of distance travelled by the first spot **135a** of the first energy beam **134a** over any particular area. A smaller length L can produce a greater travel distance of the first spot **135a** over a smaller area along the direction of travel to increase the overall amount of energy imparted on the build material **118** (FIGS. **1** and **2**). Further, the frequency of oscillations **192** in the first oscillating path **190** can be tailored by adjusting the distance D between oscillations **192**. A smaller distance D results in oscillations that are more frequent which in turn, similar to a smaller length L, can produce a greater travel distance of the first spot **135a** over a smaller area along the travel direction **D1** to increase the overall amount of energy imparted on the build material **118** (FIGS. **1** and **2**). The length L and distance D parameters may be adjusted to any suitable values that produce a suitable melt pool in the build plane **128**. In some aspects of the disclosure, such as that illustrated in FIG. **3**, the oscillations **192** may be separated from one another. However, in some aspects of the disclosure, the length L may be large enough or the distance D may be small enough that the oscillations **192** may partially overlap with one another along the travel direction T.

(52) In some aspects of the disclosure, the amplitude A, length L, and distance D for each oscillation **192** can each remain constant along the first oscillating path **190**. However, in some aspects of the disclosure, one or more of the amplitude A, length L, and distance D can independently vary along the first oscillating path **190** within oscillations **192**. For example, the amplitude A may increase for certain oscillations over a certain distance to temporarily grow the size of the melt pool with respect to the first oscillating path **190**.

(53) The oscillations **192** may comprise a variety of configurations. In some aspects of the disclosure, such as that illustrated in FIG. **3**, the oscillations **192** may comprise symmetrical loops. In some aspects of the disclosure, one or more of the oscillations **192** may additionally or alternatively comprise other configurations, such as asymmetrical loops, linear or non-linear patterns, or combinations thereof. Moreover, while the first direction **D1** is illustrated as comprising a relatively linear path, it is appreciated that the first direction **D1** may additionally, or alternatively, comprise one or more non-linear portions such as bends, curves, turns, or the like.

(54) Referring now additionally to FIG. **4**, the plurality of oscillations **192** of the first oscillating path **190** can irradiate the build material **118** (FIGS. **1** and **2**) to consolidate at least a portion of the first wall **153a**. Moreover, the plurality of oscillations **192** may define at least the first side **157a** of the first wall **153a** by irradiating and consolidating the build material **118** at the edge of the first wall **153a** without the need for additional irradiation or passes from energy beams **134** (not shown). The irradiation imparted by the first spot **135a** of the first energy beam **134a** at the peak amplitude A of each oscillation **192** will thereby cause the first side **157a** and the second side **159a** of the first wall **153a** to be consolidated to define the same. That is, the amplitude A will control the maximum dimension of the melt pool, which, in turn, becomes the outer surface of the first side **157a** and the second side **159a** of the first wall **153a**. Thus, a constant amplitude A will produce a linear first side **157a** and second side **159a** of the first wall **153a**. However, a varying amplitude A will produce a wall **153a** having a first side **157a** or second side **159a** that is nonlinear such as expanding and retracting in the direction of the thickness direction DT. Such aspects of the disclosure can allow for relatively thin-walled structures to be built for three-dimensional objects **114** with higher efficiency, precision, and uniformity compared to additively manufacturing walls using non-oscillating laser paths.

(55) The body portion **163** of the first wall **153a**, i.e., the portion between the first side **157a** of the first wall **153a** and the second side **159a** of the first wall **153a**, can be consolidated using the rest of the oscillating path **190**. Such aspects of the disclosure can avoid the need for hatching (e.g., a plurality of linear passes by one or more energy beams **134**) to help speed up the manufacturing process and avoid potential solidification inconsistencies that could occur from multiple adjacent passes.

(56) In some aspects of the disclosure, the plurality of oscillations **192** may further define the second side **159a** of the first wall **153a** by irradiating and consolidating the build material **118** at the other edge of the first wall **153a** without the need for additional irradiation or passes from energy beams **134**. The irradiation imparted by the first spot **135a** of the first energy beam **134a** at the peak amplitude **A** of each oscillation **192** will thereby also cause the second side **159a** of the first wall **153a** to be consolidated to define the same. Such aspects of the disclosure can allow for a single pass of the first energy beam **134a** on the first oscillating path **190** to define the entirety of the first wall **153a**.

(57) FIG. 5 illustrates another first oscillating path **290** according to another aspect of the disclosure. The first oscillating path **290** is similar to first oscillating path **190**; therefore, like components will be identified with like numerals increased by 100, with it being understood that the description of the like parts of the first oscillating path **190** applies to the first oscillating path **290**, unless otherwise noted. As illustrated in FIG. 5, one or more dimensions of the first oscillating path **290** are varied as the first energy beam **134a** (FIG. 1) travels in the first direction **D1**. For example, as illustrated in FIG. 5, the amplitude **A** of the first oscillating path **290** may vary as the first oscillating path travels in the first direction **D1**. In such aspects of the disclosure, the first oscillating path may comprise at least a first amplitude **A1** at a first location and a second amplitude **A2** at a second location further along in the first direction **D1**. The second amplitude **A2** can be greater than the first amplitude **A1** such that a thickness **T** of the first wall **253a** between the first side **257a** and the second side **259a** can increase along a first portion of the first wall **253a**. Depending on the design of the first wall **253a** and the three-dimensional object **114**, the amplitude **A** can vary (e.g., increase or decrease) at a single location or at a plurality of locations. Moreover, the amplitude **A** may steadily vary (e.g., a tapered increase or tapered decrease) or may quickly vary (e.g., an immediate increase or immediate decrease), or a combination thereof as the first oscillating path **190** extends in the first direction **D1**. Further, a power of the first energy beam can be adjusted when the plurality of oscillations is varied, such as by increasing or decreasing the power when the amplitude **A** is increased or decreased.

(58) The first energy beam **134a** (FIG. 1) may comprise a variety of beam parameters along the first oscillating path such as, but not limited to, power, spot size, focus depth, travel speed, and other beam parameters for sufficient consolidation of build material **118**. Similar to the amplitude **A**, the beam parameters of the first energy beam **134a** may remain constant throughout the entire oscillating path **190**, or may vary in whole or in part throughout the oscillating path **190**.

(59) FIG. 6 illustrates another first oscillating path **390** according to another aspect of the disclosure. The first oscillating path **390** is similar to first oscillating path **190**; therefore, like components will be identified with like numerals increased by 100, with it being understood that the description of the like parts of the first oscillating path **190** applies to the first oscillating path **290**, unless otherwise noted. As illustrated in FIG. 6, the first wall **353a** may be consolidated using additional irradiation such as additional passes by the first energy beam **134a** (FIG. 1) or additional passes by the second energy beam **134b** (FIG. 1). For example, with reference to FIG. 6, in some aspects of the disclosure, a second build plane region **152b** (FIG. 1) may be irradiated along a second oscillating path **210** in a second direction **D2** that is opposite the first direction **D1** to consolidate the first wall **353a**. The second oscillating path **210** can comprise a plurality of oscillations **212** that define the second side **359a** of the first wall **353a**, which is opposite the first side **357a** of the first wall **352a** to collectively define the thickness **T** of the first wall **352a**. As such, the first oscillating path **390** and the second oscillating path **210** may traverse adjacent and substantially parallel to one another to consolidate the first wall **353a**. While exemplary illustration is made to the first energy beam **134a** and the second energy beam **134b**, it is appreciated that any number of energy beams **134** may additionally or alternatively be utilized for irradiating the build plane **128** in any number of regions.

(60) In some aspects of the disclosure, the first energy beam **134a** may travel both the first

oscillating path **390** and the second oscillating path **210**. Alternatively, in some aspects of the disclosure, the first energy beam **134a** may travel the first oscillating path **390** and the second energy beam **134b** may travel the second oscillating path **210**. Moreover, the first oscillating path **390** and the second oscillating path **210** may generally comprise the same dimensions such as the same amplitude A, length L, or distance D (FIG. 3). In some aspects of the disclosure, one or more of the dimensions may vary between the first oscillating path **390** and the second oscillating path **210**. Furthermore, the parameters of the energy beam **134** used for the first oscillating path **390** and the second oscillating path **210** may be similar to one another, dissimilar to one another, or a combination thereof.

(61) Referring now to FIGS. 7-10, in some aspects of the disclosure, different oscillating paths may form adjacent portions parts of the same wall, wherein the different oscillating paths meet in an overlap space between the two adjacent portions. That is, for example, to form a wall running from right to left, one oscillating path forms the left half of the wall, another oscillating path forms the right half of the wall, and the two oscillating paths meet in an interlace region between the left half and right half of the wall. Exemplary aspects of the disclosure will be presented with reference to the individual figures, wherein like components will be identified with like numerals generally increased by 100, and with it being understood that the description of the like parts described herein carries forward unless otherwise noted.

(62) FIG. 7 illustrates a second build plane region **452b** may be irradiated along a second oscillating path **491** in a second direction D2 that is opposite the first direction D1 to consolidate an extension **471** the first wall **453a**. The first wall **453a** and the extension **471** can overlap **455** in an interlace region **454**.

(63) The interlace region **454** can be consolidated using a combination of the first energy beam **434a** and the second energy beam **434b** to help promote build quality of the three-dimensional object **114** (FIG. 1) between the first build plane region **452a** and the second build plane regions **452b** despite the use of multiple energy beams **134** (FIG. 1). In combination, the first build plane region **452a** (irradiated by the first energy beam **434a**), the second build plane region **452b** (irradiated by the second energy beam **434b**), and the interlace region **454** (irradiated by both the first energy beam **434a** and the second energy beam **434b**) can comprise a variety of relative configurations.

(64) For instance, the first oscillating path **490** of the first energy beam **134a** and the second oscillating path **491** of the second energy beam **434b** may combine and overlap **455** to irradiate the interlace region **454**. The overlap **455** refers to the portion of the build plane **128** (FIG. 1) where both the first spot **435a** of the first energy beam **434a** and the second spot **435b** of the second energy beam **434b** irradiate the same build material **118** (FIG. 1). By at least partially overlapping the first oscillating path **490** and the second oscillating path **491**, the interlace region **454** may be irradiated and consolidated in a more efficient manner and mitigate potential misalignment between the first energy beam **434a** and the second energy beam **434b**, such as by forming a larger common melt pool in the build material **118** (FIG. 1).

(65) The interlace region **454** can comprise a variety of relative configurations for overlapping, tapering, or otherwise joining the first wall **453a** with the extension **471**. For example, in some aspects of the disclosure, such as that illustrated in FIG. 7, the first oscillating path **490** and the second oscillating path **491** may overlap **455** for a defined distance. The distance and amount of overlap may depend on, for example, the configurations of the first oscillating path **490** and the second oscillating path **491**, or the parameters of the first energy beam **434a** and the second energy beam **434b**.

(66) FIG. 8 illustrates another embodiment of a first oscillating path **590** and a second oscillating path **591** forming a first wall **553a**. In some aspects of the disclosure, such as that illustrated in FIG. 8, the amplitude A for at least one of the plurality of oscillations **592**, **593** for the first oscillating path **590** and the second oscillating path **591** may vary in the interlace region **554**. For

example, the amplitude A may increase for the oscillations **592** of the first oscillating path **590** or the oscillations **593** of the second oscillating path **591** in the interlace region **554**. As a result, the first wall **553a** and the extension **571** will have a larger thickness T where they meet.

(67) FIG. **9** illustrates a first wall **653a**, wherein the first plurality of oscillations **692** of the first oscillating path **690** may taper in the interlace region **654** such that the first oscillating path **190** reduces in amplitude A as it extends towards the extension **671** until stopping or diminishing. Alternatively, or additionally, the second plurality of oscillations **693** of the second oscillating path **691** may taper in the interlace region **654** such that the second oscillating path **691** reduces in amplitude A as it extends towards the first wall **653a** until stopping or diminishing.

(68) The tapering of the first oscillating path **690** or the second oscillating path **691** may be achieved with or without overlap. For example, with reference to FIG. **9**, the oscillations **692** of the first oscillating path **690** may taper towards the first side **657a** or the oscillations **693** of the second oscillating path **691** may taper towards the second side **659a**. In such aspects of the disclosure, the first oscillating path **690** and the second oscillating path **691** may consolidate build material **118** (FIG. **1**) in the interlace region **654** without overlap.

(69) FIG. **10** illustrates a first wall **753a**, wherein the oscillations **792** of the first oscillating path **790** may taper towards the internal portion of the three-dimensional object **114** (i.e., between the first side **757a** and the second side **759a**) or the oscillations **793** of the second oscillating path **791** may taper towards the internal portion of the three-dimensional object **114** (FIG. **1**). In such aspects of the disclosure, the first oscillating path **790** and the second oscillating path **791** may consolidate build material **118** (FIG. **1**) in the interlace region **754** with overlap **755**.

(70) It is appreciated that while certain variations and permutations are presented herein with respect to the consolidation of the first wall **153a**, the extension **171**, and the interlace region **154**, these examples are not intended to be exclusive and other implementations may further be realized within the scope of this disclosure.

(71) Referring additionally to FIG. **11**, in some aspects of the disclosure, the three-dimensional object **114** may have a plurality of walls **853** that cross at one or more intersections **850**. While the three-dimensional object **814** presented in FIG. **11** illustrates the plurality of walls **853**, it is appreciated that any of the earlier series numbering and aspects of the disclosure (**153**, **253**, **353**, **453**, **552**, **653**, **753**) can be included. As illustrated in FIG. **11**, the three-dimensional object **814** may comprise a network of walls **853** that intersect at a plurality of intersections **850**, wherein one or more of the walls **853** can be consolidated using the methods and systems disclosed herein. Such aspects of the disclosure can facilitate lattice-like support structures or represent a cross section of an interleaved thin-walled structure such as heat exchanges, flow chambers, or other design configurations suitable for additive manufacturing.

(72) As illustrated in FIG. **11**, for instance, the three-dimensional object **814** may comprise a first wall **853a**, a second wall **853b**, a third wall **853c**, a fourth wall **853d**, a fifth wall **853e**, a sixth wall **853f**, and a seventh wall **853g**. The plurality of walls may comprise distinct sets of walls **853** that are parallel such as where the first wall **853a** and the second wall **853b** form a first parallel set, the third wall **853c** and the fourth wall **853d** form a second parallel set, and the fifth wall **853e**, the sixth wall **853f**, and the seventh wall **853g** form a third parallel set. While FIG. **11** illustrates seven different walls **153** in three parallel sets, it is appreciated that the three-dimensional object **814** may comprise any number of walls **853** having any suitable configuration, with or without parallel orientations.

(73) One or more of the walls **853** may comprise oscillations **892** such as via the first oscillating path **890**, and the second oscillating path **891**. For example, the first oscillating path **890** and the second oscillating path **891** may generally comprise the same dimensions such as the same amplitude A, length L, or distance D (FIG. **3**). In some aspects of the disclosure, one or more of the dimensions may vary between the first oscillating path **890** and the second oscillating path **891**. Furthermore, the parameters of the energy beam used for the first oscillating path **890** and the

second oscillating path **891** may be similar to one another, dissimilar to one another, or a combination thereof. For example, in some aspects of the disclosure, the amplitude A of the oscillations **192** may increase for one or more of, or even each of, the walls **853** at the intersections **850**. Such aspects of the disclosure may increase rigidity of the plurality of walls **853** within the three-dimensional object **814**. Alternatively, or additionally, in some aspects of the disclosure, the amplitude A of the oscillations **892** may decrease for one or more of, or even each of, the walls **853** at the intersections **850**.

(74) Referring now additionally to FIG. **12**, a method **300** is illustrated for additively manufacturing a three-dimensional object **114**. While reference of the method **300** is generally made hereinafter to the exemplary aspects of the disclosure illustrated in FIGS. **2-4**, it is appreciated that the method **300** also applies to the exemplary aspects of the disclosure included in FIGS. **5-11**. The method **300** generally comprises irradiating a first build plane region **152a** using a first energy beam **134a** travelling along a first oscillating path **190** in a first direction D1 in step **310** to consolidate a first wall **153a**, wherein a build material **118** adjacent a first side **157a** of the first wall **153a** and build material **118** adjacent a second side **159a** of the first wall **153a**, opposite the first side **157a** of the first wall **153a**, remains unconsolidated. The first oscillating path **190** can include a first plurality of oscillations **192** that define at least a first side **157a** of the first wall **153a** and a thickness T of the first wall **153a** in a thickness direction DT, perpendicular to the first direction D1, can be greater than a beam diameter DB of the first energy beam **134a**.

(75) As discussed above, the three-dimensional object **114** may comprise a variety of relative parameters and configurations with respect to one or more walls **153** and their formation.

(76) For example, the method **300** may include irradiating a second build plane region **152b** along a second oscillating path **210** in a second direction D2 opposite the first direction D1 in step **320** to consolidate the first wall **153a**. The second oscillating path **210** can include a second plurality of oscillations **212** that define a second side **159a** of the first wall **153a** opposite the first side **157a** of the first wall **153a**.

(77) In some aspects of the disclosure, such as that illustrated in FIG. **7**, the method **300** may include irradiating a second build plane region **452b** along a second oscillating path **491** in a second direction D2 opposite the first direction D1 in step **330** to consolidate an extension **471** of the first wall **453a** in the first direction D1. The second oscillating path **491** can include a second plurality of oscillations **493**. Moreover, the first plurality of oscillations **492** and the second plurality of oscillations **493** can overlap **455** in an interlace region **454**.

(78) In some aspects of the disclosure, such as that the method **300** may include irradiating a second build plane region **152b** along a second oscillating path **210** in a second direction D2 or third direction D3 different than the first direction D1 in step **340** to consolidate a second wall **853b**, wherein the first wall **853a** and the second wall **853b** intersect at an intersection **850**.

(79) Now turning to FIG. **13**, a control system **104** is illustrated. The control system **104** may be configured to perform one or more control operations associated with an additive manufacturing system **100** or an additive manufacturing machine **102**. The control operations may include one or more parts of method **300** disclosed herein, or one or more operations to otherwise consolidate the three-dimensional object **114** in accordance with the disclosure presented herein.

(80) For example, the control operations can include irradiating a first build plane region **152a** using a first energy beam **134a** travelling along a first oscillating path **190** in a first direction D1 to consolidate a first wall **153a**, wherein a build material **118** adjacent a first side **157a** of the first wall **153a** and build material **118** adjacent a second side **159a** of the first wall **153a**, opposite the first side **157a** of the first wall **153a**, remains unconsolidated. The first oscillating path **190** can include a first plurality of oscillations **192** that define at least a first side **157a** of the first wall **153a** and a thickness T of the first wall **153a** in a thickness direction DT, perpendicular to the first direction D1, can be greater than a beam diameter DB of the first energy beam **134a**.

(81) As shown in FIG. **13**, a control system **104** may include one or more computing devices **900**.

The one or more computing devices **900** may be communicatively coupled with an additive manufacturing machine **102** or additive manufacturing system **100**. For example, a computing device **900** may be communicatively coupled with one or more controllable components **902** of an additive manufacturing machine **102**, such as one or more controllable components **902** associated with an energy beam system **132**, an imaging system **158**, an inertization system **160**, or a positioning system **170**. Additionally, or in the alternative, a computing device **900** may be communicatively coupled with a management system **106** or a user interface **108**. The one or more computing devices **900** may be located locally or remotely relative to an additive manufacturing machine **102**.

(82) The one or more computing devices **900** may include one or more control modules **904** configured to cause the computing device **900** to perform one or more control operations. The one or more control modules **904** may include one or more irradiation control modules **800**. The one or more control modules **904** may include control logic executable to provide control commands configured to control the one or more controllable components **902** associated with an additive manufacturing machine **102**.

(83) The one or more computing devices **900** may include one or more processors **906** and one or more memory devices **908**. The one or more processors **906** may include any suitable processing device, such as a microprocessor, microcontroller, integrated circuit, logic device, or other suitable processing device. The one or more memory devices **908** may include one or more computer-readable media, including but not limited to non-transitory computer-readable media, RAM, ROM, hard drives, flash drives, or other memory devices **908**. The one or more control modules **904** may be implemented at least in part by the one or more processors **906** or the one or more memory devices **908**.

(84) As used herein, the terms “processor” and “computer” and related terms, such as “processing device” and “computing device”, are not limited to just those integrated circuits referred to in the art as a computer, but broadly refers to a microcontroller, a microcomputer, a programmable logic controller (PLC), an application specific integrated circuit, and other programmable circuits, and these terms are used interchangeably herein. A memory device **908** may include, but is not limited to, a non-transitory computer-readable medium, such as a random-access memory (RAM), and computer-readable nonvolatile media, such as hard drives, flash memory, and other memory devices. Alternatively, a floppy disk, a compact disc-read only memory (CD-ROM), a magneto-optical disk (MOD), or a digital versatile disc (DVD) may also be used.

(85) As used herein, the term “non-transitory computer-readable medium” is intended to be representative of any tangible computer-based device implemented in any method or technology for short-term and long-term storage of information, such as, computer-readable instructions, data structures, program modules and sub-modules, or other data in any device. The methods described herein may be encoded as executable instructions embodied in a tangible, non-transitory, computer readable media, including, without limitation, a storage device or a memory device. Such instructions, when executed by a processor, cause the processor to perform at least a portion of the methods described herein. Moreover, as used herein, the term “non-transitory computer-readable medium” includes all tangible, computer-readable media, including, without limitation, non-transitory computer storage devices, including, without limitation, volatile and nonvolatile media, and removable and non-removable media such as a firmware, physical and virtual storage, CD-ROMs, DVDs, and any other digital source such as a network or the Internet, as well as yet to be developed digital means, with the sole exception being a transitory, propagating signal.

(86) The one or more memory devices **908** may store information accessible by the one or more processors **906**, including computer-executable instructions **910** that can be executed by the one or more processors **906**. The computer-executable instructions **910** may include any set of instructions which when executed by the one or more processors **906** cause the one or more processors **906** to perform operations, including calibration operations, or additive manufacturing operations. For

example, the computer-executable instructions **910** may be configured to cause the one or more processors **906** to perform one or more of the steps presented in method **300** (FIG. **12**) or otherwise presented in this disclosure. In some aspects of the disclosure, the computer-executable instructions **910** may be configured to cause the one or more processors **906** to irradiate the first build plane region **152a** using a first energy beam **134a** travelling along a first oscillating path **190** in a first direction **D1** to consolidate a first wall **153a**, wherein a build material **118** adjacent a first side **157a** of the first wall **153a** and build material **118** adjacent a second side **159a** of the first wall **153a**, opposite the first side **157a** of the first wall **153a**, remains unconsolidated. The first oscillating path **190** can include a first plurality of oscillations **192** that define at least a first side **157a** of the first wall **153a** and a thickness **T** of the first wall **153a** in a thickness direction **DT**, perpendicular to the first direction **D1**, can be greater than a beam diameter **DB** of the first energy beam **134a**.

(87) The memory devices **908** may store a plurality of data **912** accessible by the one or more processors **906**. The data **912** may be past, current, real-time, or a combination thereof. The data **912** may be stored in a data library **914**. As examples, the data **912** may include data **912** associated with or generated by an additive manufacturing system **100** or an additive manufacturing machine **102**, including data **912** associated with or generated by a computing device **900**, an additive manufacturing machine **102**, a management system **106**, or a user interface **108**. The data **912** may also include other data sets, parameters, outputs, information, associated with an additive manufacturing system **100** or an additive manufacturing machine **102**.

(88) The one or more computing devices **900** may also include a communication interface **916** configured to communicate with various nodes on a communication network **918** via communication lines **920** that are wired or wireless. The communication interface **916** may include any suitable components for interfacing with one or more network(s), including for example, transmitters, receivers, ports, controllers, antennas, or other suitable components. The communication interface **916** may allow the one or more computing devices **900** to communicate with various nodes associated with the additive manufacturing machine **102**, the management system **106**, or a user interface **108**. The communication network **918** may include, for example, a local area network (LAN), a wide area network (WAN), satellite communication (SATCOM) network, very high frequency (VHF) network, a high frequency (HF) network, a wireless fidelity (Wi-Fi) network, a worldwide interoperability for microwave access (WiMAX) network, a gatelink network, or any other communication network **918** for transmitting messages to or from the computing device **900** across the communication lines **920**. The communication lines **920** of communication network **918** may include a data bus or a combination of wired or wireless communication links.

(89) The management system **106** may include a server **922** or a data warehouse **924**. As an example, at least a portion of the data **912** may be stored in the data warehouse **924**, and the server **922** may be configured to transmit data **912** from the data warehouse **924** to the one or more computing device **900**, or to receive data **912** from the one or more computing devices **900** and to store the received data **912** in the data warehouse **924** for further purposes. The server **922** or the data warehouse **924** may be implemented as part of the one or more computing devices **900** or as part of the management system **106**.

(90) While computer devices **900** and components thereof have been disclosed herein, it is appreciated that these specific aspects of the disclosure are not intended to be limiting and alternatives may further be realized within the scope of this disclosure.

(91) Walls **153** that are both thin and uniform can be manufactured for three-dimensional objects **114** by utilizing one or more oscillating paths **190**. By utilizing a plurality of oscillations **192**, the thickness of the walls **153** can be reduced while still promoting consistent and uniform consolidation of build material without excessive irradiation.

(92) Further aspects are provided by the subject matter of the following clauses:

(93) A method of additively manufacturing a three-dimensional object, the method comprising

irradiating a first build plane region using a first energy beam defining a beam diameter, the first energy beam travelling along a first oscillating path in a first direction to consolidate a first wall defining a thickness perpendicular to the first direction, wherein a build material adjacent a first side of the first wall and the build material adjacent a second side of the first wall, opposite the first side of the first wall, remains unconsolidated, and wherein the thickness of the first wall is greater than the beam diameter.

(94) The method of any clause herein, wherein the first oscillating path comprises a first plurality of oscillations, and wherein the thickness of the first wall is between two times and five times the beam diameter of the first energy beam.

(95) The method of any clause herein, wherein an amplitude of the first plurality of oscillations is varied along the first oscillating path.

(96) The method of any clause herein, wherein a power of the first energy beam is adjusted when the first plurality of oscillations is varied.

(97) The method of any clause herein, wherein the first oscillating path comprises a first plurality of oscillations that define at least the first side of the first wall.

(98) The method of any clause herein, further comprising irradiating a second build plane region along a second oscillating path in a second direction opposite the first direction to consolidate the first wall, wherein the first oscillating path comprises a first plurality of oscillations that define the first side of the first wall, and wherein the second oscillating path comprises a second plurality of oscillations that define the second side of the first wall.

(99) The method of any clause herein, wherein the first energy beam irradiates the second build plane region along the second oscillating path.

(100) The method of any clause herein, further comprising irradiating a second build plane region along a second oscillating path in a second direction opposite the first direction to consolidate an extension of the first wall in the first direction, wherein the second oscillating path comprises a second plurality of oscillations, wherein the first oscillating path comprises a first plurality of oscillations, and wherein the first plurality of oscillations and the second plurality of oscillations overlap in an interlace region.

(101) The method of any clause herein, wherein the first plurality of oscillations and the second plurality of oscillations taper in the interlace region.

(102) The method of any clause herein, wherein an amplitude of at least one of the first plurality of oscillations and the second plurality of oscillations varies in the interlace region.

(103) The method of any clause herein, wherein a second energy beam irradiates the second build plane region along the second oscillating path.

(104) The method of any clause herein, further comprising irradiating a second build plane region along a second oscillating path comprising a second plurality of oscillations in a second direction different than the first direction to consolidate a second wall, wherein the first wall and the second wall intersect at an intersection.

(105) The method of any clause herein wherein the first oscillating path comprises a first plurality of oscillations, and wherein an amplitude of at least one of the first plurality of oscillations and the second plurality of oscillations increases at the intersection.

(106) The method of any clause herein, wherein a second energy beam irradiates the second build plane region along the second oscillating path.

(107) An additive manufacturing system for additively manufacturing a three-dimensional object, the additive manufacturing system comprising, a first irradiation device configured to generate a first energy beam; a first optical assembly configured to direct the first energy beam; and a control system configured to perform one or more control operations associated with the additive manufacturing system, wherein the one or more control operations comprise irradiating a first build plane region using the first energy beam defining a beam diameter, the first energy beam travelling along a first oscillating path in a first direction to consolidate a first wall defining a thickness

perpendicular to the first direction, wherein a build material adjacent a first side of the first wall and adjacent a second side of the first wall, opposite the first side of the first wall, remains unconsolidated; and wherein the thickness of the first wall is greater than the beam diameter of the first energy beam.

(108) The additive manufacturing system of any clause herein, wherein the one or more control operations further comprise irradiating a second build plane region along a second oscillating path in a second direction opposite the first direction to consolidate the first wall, wherein the first oscillating path comprises a first plurality of oscillations, wherein the second oscillating path comprises a second plurality of oscillations that define the second side of the first wall. The additive manufacturing system of claim 15, wherein the one or more control operations further comprise irradiating a second build plane region along a second oscillating path in a second direction opposite the first direction to consolidate an extension of the first wall in the first direction, wherein the first oscillating path comprises a first plurality of oscillations, wherein the second oscillating path comprises a second plurality of oscillations, and wherein the first plurality of oscillations and the second plurality of oscillations overlap in an interlace region.

(109) The additive manufacturing system of any clause herein, wherein the one or more control operations further comprise irradiating a second build plane region along a second oscillating path in a third direction different than the first direction to consolidate a second wall, wherein the first wall and the second wall intersect at an intersection.

(110) The additive manufacturing system of any clause herein, further comprising a second irradiation device configured to generate a second energy beam; and a second optical assembly configured to direct the second energy beam, wherein the second energy beam irradiates the second build plane region along the second oscillating path.

(111) A three-dimensional object manufactured by a method comprising irradiating a first build plane region using a first energy beam defining a beam diameter, the first energy beam travelling along a first oscillating path in a first direction to consolidate a first wall defining a thickness perpendicular to the first direction, wherein a build material adjacent a first side of the first wall and the build material adjacent a second side of the first wall, opposite the first side of the first wall, remains unconsolidated, and wherein the thickness of the first wall is greater than the beam diameter.

(112) This written description uses examples to disclose the present disclosure, including the best mode, and also to enable any person skilled in the art to practice the disclosure, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the disclosure is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they include structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A method of additively manufacturing a three-dimensional object, the method comprising: irradiating a first build plane region using a first energy beam defining a beam diameter, the first energy beam travelling along a first oscillating path having a varying first amplitude in a first direction to consolidate a first wall defining a thickness perpendicular to the first direction, wherein a build material adjacent a first side of the first wall and the build material adjacent a second side of the first wall, opposite the first side of the first wall, remains unconsolidated, and irradiating a second build plane region along a second oscillating path having a varying second amplitude in a second direction opposite the first direction to consolidate the first wall, wherein a portion of the first oscillating path travelling in the first direction overlaps a portion of the second oscillating path

travelling in the second direction opposite the first direction, wherein the first oscillating path includes a plurality of first oscillations that bend around and cross itself and the second oscillating path includes a plurality of second oscillations that bend around and cross itself, and wherein the plurality of first oscillations also bend around and cross the plurality of second oscillations, and wherein the thickness of the first wall is greater than the beam diameter, wherein the plurality of first oscillations and the plurality of second oscillations overlap in an interlace region, and wherein the varying first amplitude increases in the interlace region and the varying second amplitude increases in the interlace region.

2. The method of claim 1, wherein the thickness of the first wall is between two times and five times the beam diameter of the first energy beam.

3. The method of claim 2, wherein the varying first amplitude is different than the varying second amplitude.

4. The method of claim 3, wherein a power of the first energy beam is adjusted when the plurality of first oscillations is varied.

5. The method of claim 1, wherein the first oscillating path comprises the plurality of first oscillations that define at least the first side of the first wall.

6. The method of claim 1, wherein the first oscillating path comprises the plurality of first oscillations that define the first side of the first wall, and wherein the second oscillating path comprises the plurality of second oscillations that define the second side of the first wall, and wherein the first oscillating path is different than the second oscillating path.

7. The method of claim 6, wherein the first energy beam irradiates the second build plane region along the second oscillating path.

8. The method of claim 1, wherein irradiating the second build plane region along the second oscillating path consolidates an extension of the first wall in the first direction, wherein the first oscillating path comprises the plurality of first oscillations, wherein the second oscillating path comprises the plurality of second oscillations, and wherein the plurality of first oscillations and the plurality of second oscillations overlap in the interlace region.

9. The method of claim 8, wherein the plurality of first oscillations and the plurality of second oscillations taper in the interlace region, and wherein the plurality of first oscillations taper until stopping.

10. The method of claim 8, wherein the varying first amplitude and the varying second amplitude vary in the interlace region.

11. The method of claim 8, wherein a second energy beam irradiates the second build plane region along the second oscillating path.

12. The method of claim 1, wherein irradiating the second build plane region along the second oscillating path comprising the plurality of second oscillations in the second direction different than the first direction consolidates a second wall, wherein the first wall and the second wall intersect at an intersection.

13. The method of claim 12, wherein the varying first amplitude and the varying second amplitude increase at the intersection.

14. The method of claim 12, wherein a second energy beam irradiates the second build plane region along the second oscillating path.

15. The method of claim 1, wherein the method further comprises: forming a network of walls that intersect at a plurality of intersections.

16. The method of claim 15, wherein the network of walls comprise the first wall, a second wall, a third wall, a fourth wall, a fifth wall, and a sixth wall, wherein the first wall and the second wall form a first parallel set, wherein the third wall and the fourth wall form a second parallel set, and wherein the fifth wall and the sixth wall form a third parallel set.
